

Filipe C. Thewes

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University of Luxembourg

Scientific positions and education:

01-02/2027	Visiting researcher , University of Cambridge DAMPT - Group of Michael Cates
2025-2027	Postdoctoral researcher , University of Luxembourg Supervisor: Etienne Fodor
2024-2025	Postdoctoral researcher , Max-Planck Institute for Dynamics and Self-Organization Supervisor: David Zwicker
2020-2024	PhD in Physics , University of Göttingen Thermodynamics and kinetics of multicomponent mixtures (summa cum laude) Supervisor: Peter Sollich
2018-2019	MSc in Physics , Universidade Federal do Rio Grande do Sul, Brazil Supervisor: Heitor C. M. Fernandes
2013-2018	BSc in Physics , Universidade Federal do Rio Grande do Sul, Brazil

Grants and scholarships

2025	University of Luxembourg RISE Fellowship (2500€)
2024	Newton Gateway to Mathematics travel grant (340£)
2023	StatPhys28 travel grant (40k JPY)
2023	Max-Planck IMPRS travel grant (750€)
2020-2024	CNPq PhD scholarship – declined (~201k BRL)
2018-2019	CNPq Masters scholarship (~36k BRL)
2017-2018	CAPES undergraduate scholarship (~9.6k BRL)

Teaching experience

2026	Lecturer : Non-equilibrium soft and active matter (masters, University of Luxembourg)
2022-2023	Tutor : Classical Field Theory (bachelors, University of Göttingen)
2021-2022	Tutor : Thermodynamics and Statistical Mechanics (bachelors, University of Göttingen)
2015-2016	Tutor : Computational Methods in Physics (bachelors, Uni. Fed. do Rio Grande do Sul)
2010	Volunteer tutor : Introductory course to informatics (INEVAM orphanage, Brazil)

Conference presentations

Nonlinear Dynamics and Emergence Across Scales: Active, Chaotic, and Complex Systems (Roscoff, France)
10th Edwards Symposium (Cambridge, UK)
Journées de Physique Statistique (Paris, France)
8th Edwards Symposium (Cambridge, UK)
StatPhys28 (Tokyo, Japan)
DPG Spring meeting (Dresden, Germany)
DPG Spring meeting (Berlin, Germany)
DPG spring meeting (Regensburg, Germany)
Hi-noon seminar series (Göttingen, Germany)

Workshops

Driven and active amorphous matter (Göttingen, Germany).
Random Matrices, Random Graphs and Statistical Physics for Machine Learning and Inference (Trieste, Italy).
Biological condensates: cellular mechanisms governed by phase transitions (Cambridge, UK).

Reviewing services

Reviewer for *ACS Omega*, *Soft Matter*, *Physical Review E*, and *Physical Review R*

Supervision

Co-supervision of Ph.D.: Adhvik Jagannathan

Master thesis: Adhvik Jagannathan

Bachelor thesis: Theo Richter, Janik Florczak.

Pre-prints

- [1] **Speed up of passive tracers in mixtures with active chemical reactions.**
FC Thewes, C Luo, & D Zwicker (2026). arXiv:2609.29416

Publications

- [6] **Phase separation with nonlocal interactions.**
FC Thewes, Y Qiang, OW Paulin, & D Zwicker. (2026). PRR (Vol. 8, Issue 2)
- [5] **Ensemble inequivalence in the design of mixtures with super-Gibbs phase coexistence.**
FC Thewes, & P Sollich. (2025). PRE (Vol. 112, Issue 2).
- [4] **Mobility-induced kinetic effects in multicomponent mixtures.**
FC Thewes, M Krüger, & P Sollich. (2024). Europhys. Lett. (Vol. 147, Issue 2)
- [3] **Nonequilibrium mixture dynamics: A model for mobilities and its consequences.**
M Akaberian, FC Thewes, P Sollich, & M Krüger. (2023). J. Chem. Phys. (Vol. 158, Issue 21)
- [2] **Composition Dependent Instabilities in Mixtures with Many Components.**
FC Thewes, M Krüger, & P Sollich. (2023). PRL (Vol. 131, Issue 5).
- [1] **Phase transitions in hard-core lattice gases on the honeycomb lattice.**
FC Thewes, & HCM Fernandes. (2020). PRE (Vol. 101, Issue 6)

References

Prof. Dr. Peter Sollich, Prof. Dr. Étienne Fodor, Prof. Dr. Matthias Krüger, Dr. David Zwicker.